

Anshul Panigrahi

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EDUCATION

Vellore Institute of Technology

BTech in Information Technology | CGPA: 8.48/10

- Relevant Coursework: Machine Learning, Distributed Systems, Cloud Architecture Design, Computer Networks, Database Management Systems

Vellore, Tamil Nadu

Aug. 2023 – Jul. 2027

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow Lite, scikit-learn, Ollama, Generative AI, Computer Vision, Model Quantization, Synthetic Data Generation

Agentic AI: Multi-agent Orchestration, N8N, Claude Code, OpenAI Codex, Cursor Agent SDK, OpenClaw, Hermes

Data & Infrastructure: Apache Kafka, Apache Spark, Cassandra, Lambda Architecture, Edge AI, Embedded Systems, Docker

Languages & Frameworks: Python, FastAPI, Flask, Next.js, SQL, REST APIs, Git, Linux

PROJECTS

Real-Time Predictive Maintenance | *Python, Kafka, Spark, PyTorch, Next.js*

Patent Application Filed, Feb. 2026

- Built a streaming data pipeline across Kafka, Spark, Cassandra, and PyTorch monitoring 5,000 factory machines; trained a Random Forest classifier on the UCI AI4I 2020 dataset with an 80/20 split, reaching 98% accuracy with an F1 score of 0.96 and AUC of 0.99.
- Designed an adaptive edge-cloud routing system using an Isolation Forest model for local anomaly detection, offloading to cloud only when needed via a self-tuning controller, keeping end-to-end latency under 12ms.
- Applied PageRank and Label Propagation on the machine connectivity graph to identify cascade failure risks; visualized all 5,000 machines live in a 3D Next.js and Three.js dashboard.

Physical-Bounded Multimodal Mode Discovery | *Python, scikit-learn, HDBSCAN*

- Discovered hidden fault patterns in NASA CMAPSS and CWRU benchmark datasets using unsupervised HDBSCAN clustering without any labeled training data, uncovering 7 and 8 viable fault regimes respectively.
- Validated clusters against physical constraints including ISO 10816 vibration limits and Joule heating bounds, achieving an 80% constraint pass rate and significantly reducing false-positive detections over a flat K-Means baseline.
- Built a physics-informed auto-labeling pipeline that generates synthetic fault samples strictly within safe operating bounds, eliminating manual annotation effort for new fault classes.

Stella | *Python, FastAPI, Mistral 7B, Ollama*

- Built an on-device AI health assistant where a locally-hosted Mistral 7B language model autonomously interprets wearable data and answers natural language health queries, with zero data leaving the device across 29 tracked metrics for 33 users.
- Implemented a Z-score sliding window anomaly detector that feeds flagged events directly into the language model as context, enabling the model to generate plain-language explanations for each anomaly flagged across 31 days of continuous monitoring.

Veri-Dose | *Python, TensorFlow Lite, Raspberry Pi*

Patent Published, Mar. 2026 | IN202641027860 A1

- Designed an offline edge AI medication dispenser using computer vision and 4-class image classification to identify pill types and prevent dispensing errors, with no internet connection or cloud inference required.
- Optimized the vision model for on-device deployment via TensorFlow Lite post-training quantization, achieving under 100ms inference latency on a Raspberry Pi without accuracy degradation.

CERTIFICATIONS

NVIDIA DLI Getting Started with Deep Learning
Oracle Cloud Infrastructure 2025 AI Foundations Associate
HackerRank Software Engineer Intern Certificate
Kaggle Intro to Machine Learning

Mar. 2026
Mar. 2026
Feb. 2026
Feb. 2026

ACHIEVEMENTS

2nd Place – IEEE CS Hackathon, real-time posture correction system
Finalist – Yantra Central Hackathon, fingerprint spoofing detection system

Sep. 2025
Feb. 2025